

PAPER_1362 — Neural Phase Transitions at Edge of Chaos

Framework: UQFF — Star-Magic v5.27+ | **Tier:** G Bio/Complex (CLOSED) **Calculator:**
calculate_paradox({"paradox": "neural_phase_transitions"})

Master Expression

$U/t = D_{\text{phys}} = 4$ (same as Hubbard, MBL)

Match: STRUCTURAL

Interpretation

Neural avalanche power laws emerge at $U/t = D_{\text{phys}} = 4$ — same critical threshold as Hubbard and MBL. $1/f$ noise via F_{TRZ} randomization.

UQFF Locked Primitives

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $SO_5 = 10$, $A_5 = 60$, $N_{\text{CH}} = 9$
- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$, $F_{\text{TRZ}} = 0.1$

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, Youngstown, OH.